

the intervals cross zero. The fresh Real-IAD C1/C2 official attempt beats CW ($\Delta = +0.0191$, 95% CI $[+0.0148, +0.0235]$) but fails SAR ($\Delta = -0.0858$, 95% CI $[-0.0928, -0.0787]$). The strict legacy `gate_e_m2_transfer_confirmed` field therefore remains false, and `gate_e_positive_transfer_confirmed` remains false until a fresh or unopened natural holdout passes both co-endpoints.

D16 natural-degradation headroom track. The D16 track moves from clean natural transfer to natural degraded samples in Real-IAD D3. Because `common_mode_filter` was opened during smoke testing and debugging, the official holdout excludes that category and uses the remaining nineteen categories. A validation-only selector chooses `score_disagreement` on the validation stress subset, routes only detected stress samples through that method, and defaults clean samples to CW. The all-test CW comparison is positive ($\Delta = +0.0624$, 95% CI $[+0.0347, +0.0896]$), and clean no-regression passes. The locked primary stress-subset endpoint is not confirmed because its paired-bootstrap interval crosses zero ($\Delta = +0.0351$, 95% CI $[-0.0276, +0.0980]$). Thus `gate_s_natural_degradation_confirmed` remains false.

Role of the new Real-IAD D3 data. The new Real-IAD D3 data is used to answer a different question from the failed clean-transfer Gate E study. It is not a tuning set, and it is not used to rewrite the definition of transfer success. Its role is to test whether the reliability-stress mechanism still has measurable headroom when degradation is natural rather than synthetically injected. The result is useful but not decisive: the full D3 test set shows a positive CW comparison, while the locked stress-subset endpoint and the SAR comparison remain insufficient. Therefore the dataset strengthens the audit and explains the remaining failure mode, but does not change the readiness tier.

Table 14: D16 Real-IAD D3 natural-degradation audit. The primary endpoint is the stress subset versus confidence-weighted mean (CW); all-test rows are supportive only.

Endpoint	Comparator	Δ AUC (95% CI)	Decision
Stress subset	CW	$+0.0351 [-0.0276, +0.0980]$	fail: CI crosses zero
Clean subset	CW	$+0.0000 [+0.0000, +0.0000]$	pass: no regression
All test	CW	$+0.0624 [+0.0347, +0.0896]$	supportive only
All test	SAR	$-0.2939 [-0.3213, -0.2665]$	fail

Algorithm 1 Patch-level PatchCore upstream detector (per modality)

Require: Train-good images $\mathcal{N}$, queries $\mathcal{Q}$, backbone blocks ℓ_2, ℓ_3 , grid G , coresets fraction c

- 1: **function** PATCHES(x)
 - 2: $f_2, f_3 \leftarrow \ell_2(x), \ell_3(x)$; locally-aware 3×3 avg-pool; resize to $G \times G$
 - 3: **return** $\{[f_2; f_3]_u : u \in G \times G\}$
 - 4: **end function**
 - 5: $\mathcal{B} \leftarrow \text{GREEDYCORESET}(\bigcup_{n \in \mathcal{N}} \text{PATCHES}(n), c|\cdot|)$
 - 6: $s_q \leftarrow \max_{p \in \text{PATCHES}(q)} \min_{b \in \mathcal{B}} \|p - b\|_2$ for $q \in \mathcal{Q}$; z-sigmoid normalise vs. train-good
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